

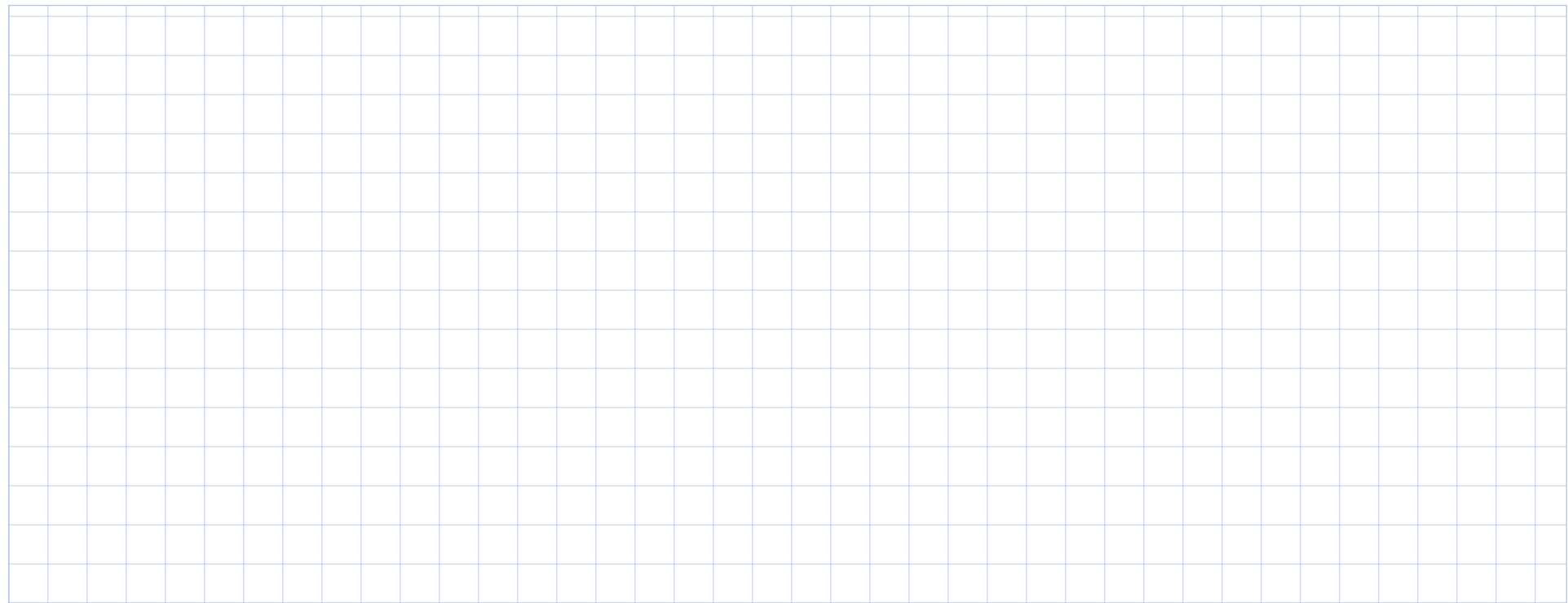
Even

f is **even** when $f(-x) = f(x)$ for every x in its domain. Its graph is symmetric about the y -axis.

Odd

f is **odd** when $f(-x) = -f(x)$ for every x in its domain. Its graph is symmetric about the origin.

To classify a formula, compute $f(-x)$, simplify completely, and compare it with both $f(x)$ and $-f(x)$. A function may be neither even nor odd.



Classify Each Function

Determine whether each function is **even**, **odd**, or **neither**. Show the algebraic test.

a. $f(x) = x^4 - 3x^2 + 1$

b. $g(x) = x^5 - 4x$

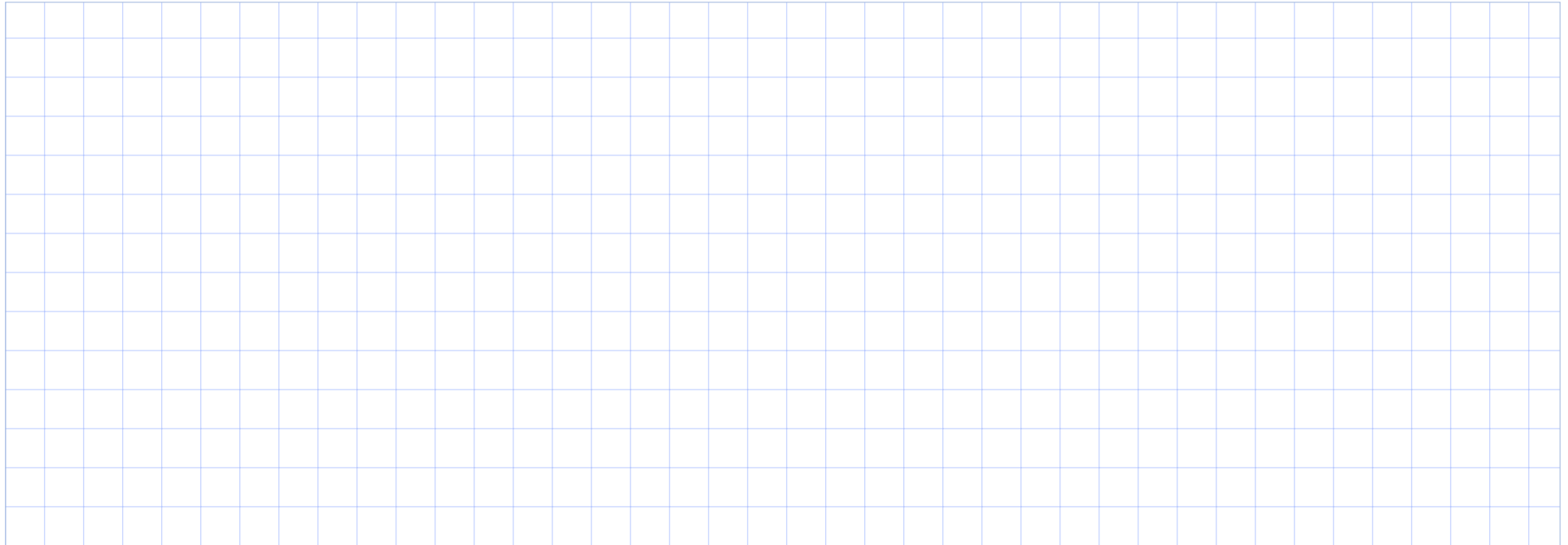
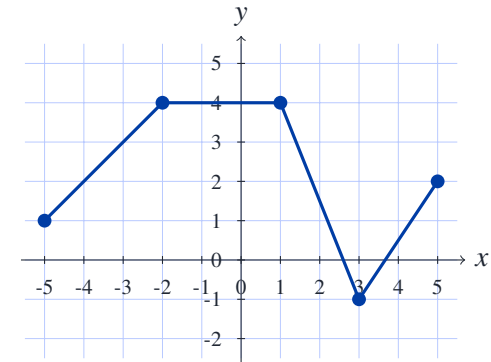
c. $h(x) = x^3 + x^2$

Where Is the Function Increasing or Decreasing?

From the graph, state the intervals on which f is:

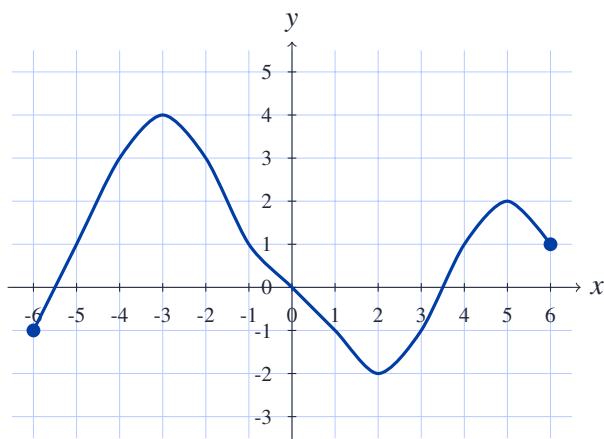
- a. increasing,
- b. decreasing,
- c. constant.

Use interval notation and do not list isolated endpoints as intervals.



Identify Local and Absolute Extrema

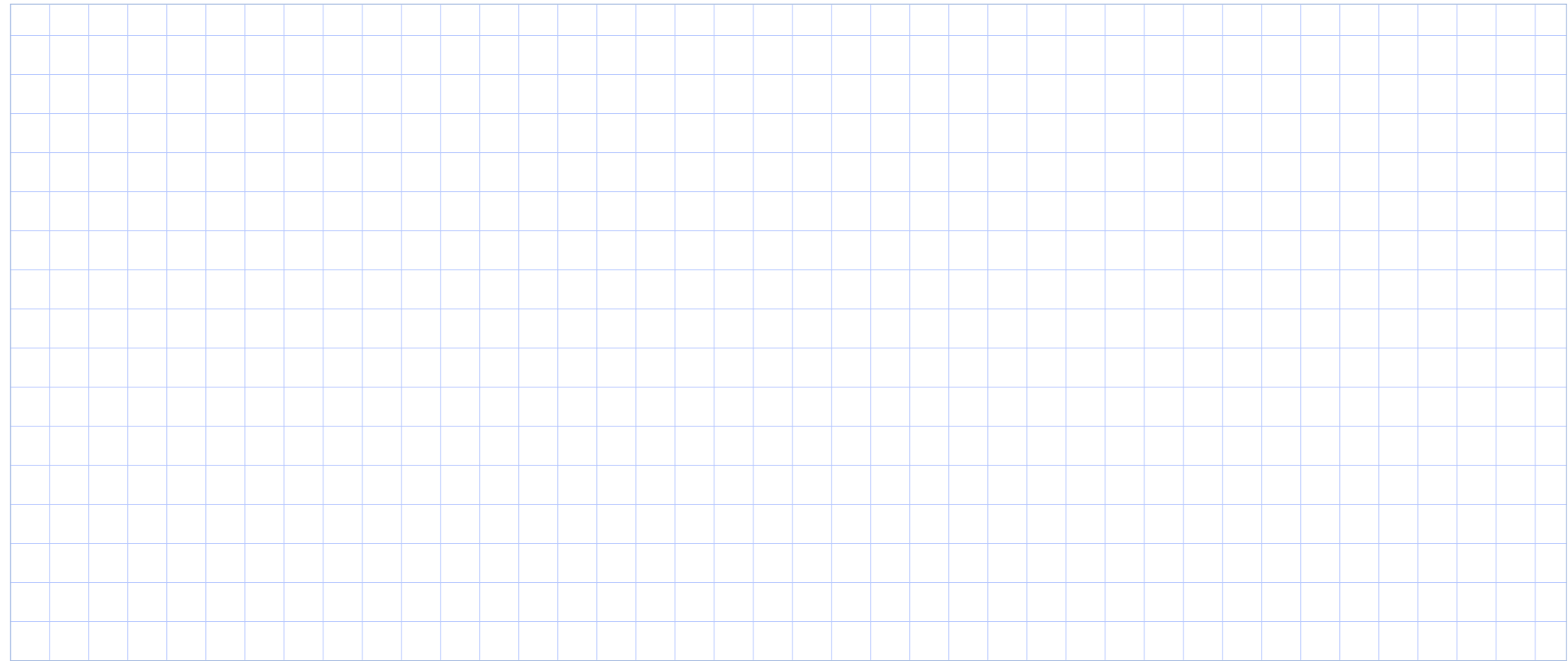
Use the graph below. List every local maximum and minimum, then identify the absolute maximum and minimum on the shown domain. Report each as a point $(x, f(x))$.



Average Rate of Change

From $x = a$ to $x = b$, the average rate of change is $\frac{f(b) - f(a)}{b - a}$. It is also the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

For $f(x) = 2x^2 - 3x + 1$, find the average rate of change from $x = -1$ to $x = 2$. Interpret the sign of your answer.



Find the Secant Line

Let $f(x) = x^2 - 4x + 5$. Find the equation of the secant line through the points on the graph with $x = 0$ and $x = 3$.

- Find both points.
- Compute the secant slope.
- Write the line in slope-intercept form.
- Check that both points satisfy your equation.

